

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2200

Farrell 138 kV
February 01, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2200. The Interconnection Customer proposes a 229.5 MW wind facility interconnecting at the Farrell 138 kV substation in the Virginia area. The request was received on October 10, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 648 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,494,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Le Gand - West Winchester 138 kV
Contingency: W Winchester - International Paper Valliant Ok 138 kV L
Loading: 122.2% (Rating: 337 MVA)
DFAX: 9.3%
Pre-project loading: 115.8%
2. Facility: Abbotsford - Le Grand 138 kV L
Contingency: International Paper Valliant Ok - West Winchester 138 kV Line
Loading: 118.2% (Rating: 370 MVA)
DFAX: 9.3%
Pre-project loading: 112.4%
3. Facility: Ocean Road - Monolith 138 kV
Contingency: International Paper Valliant Ok - Kearneyville 345 kV Line
Loading: 114.7% (Rating: 367 MVA)
DFAX: 8.0%
Pre-project loading: 109.7%
4. Facility: N Mesa - Sandalfoot 138 kV
Contingency: International Paper Valliant Ok - Kearneyville Substation 345 kV
Loading: 113.1% (Rating: 425 MVA)
DFAX: 5.4%
Pre-project loading: 110.2%
5. Facility: Celeron Substation - Sandalfoot 138 kV
Contingency: Kearneyville - International Paper Valliant Ok 345 kV
Loading: 112.2% (Rating: 442 MVA)
DFAX: 5.4%
Pre-project loading: 109.5%

6. Facility: Farrell - Depere 138 kV
Contingency: International Paper Valliant Ok - Kearneyville 345 kV
Loading: 112.2% (Rating: 233 MVA)
DFAX: 26.4%
Pre-project loading: 86.1%
7. Facility: Arcola - Twin Lakes Alte 138 kV
Contingency: International Paper Valliant Ok - Kearneyville 345 kV
Loading: 106.4% (Rating: 482 MVA)
DFAX: 5.8%
Pre-project loading: 103.6%
8. Facility: Arcola - New Road 138 kV L
Contingency: Kearneyville - International Paper Valliant Ok 345 kV
Loading: 105.7% (Rating: 453 MVA)
DFAX: 5.8%
Pre-project loading: 102.7%
9. Facility: New Rd Substation - Ag Reliant 138 kV Line
Contingency: International Paper Valliant Ok - Kearneyville Substation 345 kV
Loading: 105.3% (Rating: 440 MVA)
DFAX: 5.8%
Pre-project loading: 102.3%
10. Facility: Huntingdon District - Ag Reliant 138 kV
Contingency: Kearneyville - International Paper Valliant Ok 345 kV line
Loading: 104.2% (Rating: 401 MVA)
DFAX: 5.8%
Pre-project loading: 100.8%
11. Facility: W Winchester - Santa Rosa Energy Center 138 kV Line 1
Contingency: W Winchester - Santa Rosa Energy Ctr 138 kV
Loading: 102.0% (Rating: 445 MVA)
DFAX: 6.4%
Pre-project loading: 98.7%
12. Facility: Santa Rosa Energy Ctr - W Winchester 138 kV Cir. 2
Contingency: Santa Rosa Energy Center - W Winchester 138 kV Circuit 1
Loading: 102.0% (Rating: 445 MVA)
DFAX: 6.4%
Pre-project loading: 98.7%
13. Facility: Old Chapel - Hp Hoo Ct 138 kV
Contingency: International Paper Valliant Ok - Kearneyville 345 kV
Loading: 101.9% (Rating: 407 MVA)
DFAX: 7.2%
Pre-project loading: 97.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Le Grand - W Winchester 138 kV	Rebuild line with higher-capacity conductor	\$3,500,000	32% (\$1,103,000)
Le Grand - Abbotsford 138 kV	Rebuild line with higher-capacity conductor	\$7,008,000	32% (\$2,208,000)
Ocean Rd - Monolith 138 kV Line	Rebuild line with higher-capacity conductor	\$1,271,000	8% (\$100,000)
N Mesa - Sandalford 138 kV	Rebuild line with higher-capacity conductor	\$2,475,000	6% (\$154,000)
Celeron Substation - Sandalford 138 kV	Rebuild line with higher-capacity conductor	\$3,854,000	6% (\$240,000)
Farrell - Depere 138 kV	Rebuild line with higher-capacity conductor	\$4,697,000	30% (\$1,419,000)
Arcola - Twin Lakes Alte 138 kV	Reconductor line	\$2,198,000	6% (\$144,000)
Arcola - New Road 138 kV L	Reconductor line	\$616,000	6% (\$40,000)
New Rd Substation - Ag Reliant 138 kV Line	Reconductor line	\$1,490,000	6% (\$97,000)
Huntingdon District - Ag Reliant 138 kV	Reconductor line	\$3,290,000	6% (\$215,000)
W Winchester - Santa Rosa Energy Ctr 138 kV Ckt 1	Reconductor line	\$1,674,000	8% (\$135,000)
Santa Rosa Energy Ctr - W Winchester 138 kV Cir. 2	Reconductor line	\$2,312,000	8% (\$187,000)
Old Chapel - Hp Hood Ct 138 kV line	Reconductor line	\$3,859,000	12% (\$452,000)
	Total allocated to project		\$6,494,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.